

PHYS-467 Machine Learning for Physicists
Graded Homework n. 2: Bayesian Estimation with Monte Carlo
Due: December 3, 2021, 23:59

Instructions

- Please, upload your solution to the Moodle page of the course. No delay is accepted.
- The solution must be **submitted** on moodle: drafts will not be graded.
- The solution must consist of one notebook (.ipynb file) containing both the code and the written comments.
- The file should be called surname_name.ipynb. (e.g. Giovanni Piccioli $\mapsto$ piccioli_giovanni.ipynb). Use only characters in [a-z] (e.g. à,ç,é $\mapsto$ a,c,e) and use an underscore instead of the space.

The homework:

In this homework we will consider a variant of the spin glass card game. The difference from what we covered in the lecture is that now we have 3 types of cards in the deck. We deal one card at random to every person and call this ground truth values of cards $s^* = (s_1^*, \dots, s_N^*)$, where $s_i^* \in \{0, 1, 2\}$, $i = 1, \dots, N$. Each type of card is equally represented in the deck, i.e. $P(s_i^* = 0) = P(s_i^* = 1) = P(s_i^* = 2) = 1/3$.

The observed data is an $N \times N$ matrix Y which is generated in the following way

$$Y_{ij} = \begin{cases} \frac{1}{\sqrt{N}} (2\delta_{s_i^*, s_j^*} - 1) + \sqrt{\Delta} W_{ij} & \text{if } i < j \\ 0 & \text{if } i = j \\ Y_{ji} & \text{if } i > j \end{cases}, \quad (1)$$

where $W_{ij} \sim \mathcal{N}(0, 1)$ are all independent across pairs i, j .

1. Show that the posterior distribution of s given Y can be written as

$$P(s|Y) = \frac{1}{Z(Y, \Delta)} e^{-\frac{1}{\Delta} H(s; Y, \Delta)}; \quad H(s; Y, \Delta) = -\frac{1}{\sqrt{N}} \sum_{i < j} Y_{ij} (2\delta_{s_i, s_j} - 1), \quad (2)$$

where it is implicit that $s_i \in \{0, 1, 2\}$ for all $i = 1, \dots, N$. Note this is called the Potts model in statistical physics.

2. Generate s^* and Y according to the procedure described above. Consider parameters $N = 128$, $\Delta = 0.28$.
3. Consider a Metropolis Monte Carlo that aims to sample the posterior distribution. Consider the following moves: suppose the current state of the Markov chain is $s = (s_1, \dots, s_N)$.
 - (a) An index i is sampled uniformly from $\{1, 2, \dots, N\}$
 - (b) The proposed state is $s' = (s_1, \dots, s_{i-1}, s'_i, s_{i+1}, \dots, s_N)$ where

$$s'_i = \begin{cases} (s_i + 1) \bmod 3 & \text{with probability } 1/2 \\ (s_i + 2) \bmod 3 & \text{with probability } 1/2 \end{cases} \quad (3)$$

Derive what are the probabilities that such a proposed move is accepted. Use the general Metropolis rule we introduced in the lecture.

4. Implement the above Metropolis Monte Carlo. Initialize the Monte Carlo with state s^0 with i.i.d. coordinates each uniformly distributed in $\{0,1,2\}$. Run the Monte Carlo for $T = 10^5$ time steps. Plot the energy $H(s^t)$, $t = 0, \dots, T$ as a function of time. Around which time has your Monte Carlo equilibrated? Explain your reasoning.

Note: While implementing the algorithm you will need to compute the energy difference $H(s') - H(s)$: this computation should require $O(N)$ time.

5. Now consider the overlap between the current configuration and the ground truth $O(s, s^*) = \frac{1}{N} \sum_{i=1}^N \delta_{s_i s_i^*}$. It can be proved (you don't have to prove it) that the estimator that maximizes this overlap is $\hat{s}(Y) = (\hat{s}_1(Y), \dots, \hat{s}_N(Y))$, with

$$\hat{s}_i(Y) = \operatorname{argmax}_{k \in \{0,1,2\}} \mu(s_i = k), \quad (4)$$

where $\mu(s_i)$ is the marginal of variables i of the posterior probability distribution $P(s|Y)$. Compute $\hat{s}_i(Y)$ by estimating the marginals $\mu(s_i = k)$ from the Monte Carlo. Explain how you estimated the marginals.

6. Repeat the procedure for several values of Δ and plot the achieved overlap as a function of Δ . Notice that there is a permutational symmetry between the 3 types of cards, the Monte Carlo randomly picks one of the permutations. Consequently, plot the overlap for the permutation that gives the largest overlap $\max_{\pi} O(\pi(s), s^*)$, here π are the permutations on 3 elements. For example if $\pi = (2, 0, 1)$ and $s = (0, 0, 1, 2, 0)$, then $\pi(s) = (2, 2, 0, 1, 2)$. This is the analogue of taking the absolute value of the magnetization for Ising variables.